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null vs. undefined





JavaScript comes with two values that means "no value": `undefined` and `null`.

`undefined` means a variable has been declared but has not been assigned a value.

`null`, on the other hand, is an intentional absence of value assigned by the developer.

`undefined`

`undefined` is the default value for uninitialized variables, missing function arguments, and object properties that do not exist.

```js
let name;
console.log(name); // undefined
```

```js
function greet(message) {
  console.log(message);
}
greet(); // undefined
```

```js
const user = {};
console.log(user.age); // undefined
```

`null`

`null` represents a deliberate "no value". It is often used to clear a variable or indicate that an object reference is intentionally empty.

```js
let user = { name: "Ada" };
user = null; // intentionally cleared

console.log(user); // null
```

Key differences

Aspect	`undefined`	`null`
Meaning	Value not assigned	Intentional absence of value
Type	`"undefined"`	`"object"` (legacy bug)
Default	Given by JavaScript	Assigned by developer
`==` comparison	`undefined == null` is `true`	`null == undefined` is `true`
`===` comparison	`undefined === null` is `false`	`null === undefined` is `false`

`typeof` behavior

```js
typeof undefined; // "undefined"
typeof null; // "object" (a legacy JavaScript quirk)
```

To check for `null`, use strict equality.

```js
const value = null;
value === null; // true
```

Checking for empty values

Use strict equality when you need to distinguish between `null` and `undefined`.

```js
const a = null;
const b = undefined;

a === null; // true
b === undefined; // true
a === b; // false
```

Use loose equality when either value should be treated as empty.

```js
const value = null;

if (value == null) {
  console.log("value is null or undefined");
}
```

Nullish coalescing

The `??` operator treats only `null` and `undefined` as missing values.

It returns the right-hand side if the left-hand side is `null` or `undefined`, otherwise it returns the left-hand side.

```js
const count = 0;
console.log(count ?? 10); // 0

const name = null;
console.log(name ?? "Anonymous"); // "Anonymous"
```

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